

A TENSOR ANGLE INEQUALITY FOR PROJECTIVE NORMS AND PRESERVATION OF A GRAM DOMINATION ORDER UNDER MULTILINEAR CONTRACTIONS

REVIEW NOTE (EXTRACTED FROM THEOREM R, V2)

1. WHAT IS TO BE CHECKED

Everything takes place in finite-dimensional *real* Hilbert spaces. For $\rho \geq 0$ and $\psi \in [0, \pi]$ let

$$K(\rho, \psi) := \begin{pmatrix} 1 & \rho \cos \psi \\ \rho \cos \psi & \rho^2 \end{pmatrix}$$

be the Gram matrix of a unit vector and a vector of norm ρ at angle ψ . Angles are $\angle(a, b) = \arccos(\langle a, b \rangle / (\|a\| \|b\|))$, and $\preceq$ is the Löwner order.

Definition 1 (Order O2). A pair (F, R) is *dominated by* (ρ, χ) , with $\rho, \chi \geq 0$, if $\text{Gram}(F, R) \preceq K(\rho, \psi')$ for some $\psi' \in [0, \min(\chi, \pi)]$.

Assumptions.

- (H1) $\mu : H_1 \times \cdots \times H_m \rightarrow H$ is multilinear with $\|\mu\| := \sup\{\|\mu(x_1, \dots, x_m)\| : \|x_i\| \leq 1\} \leq 1$, $m \geq 1$.
- (H2) P is an orthogonal projector on H , $Q = I - P$, and $\eta \in (0, 1]$.
- (H3) For each i , $(F_i, R_i) \in H_i^2$ is dominated by (ρ_i, χ_i) .
- (H4) *Trajectory closure*: $\|Q\mu(R_1, \dots, R_m)\| \leq \eta \prod_i \|R_i\|$.

Output. $F := \mu(F_1, \dots, F_m)$ and $R := P\mu(R_1, \dots, R_m)$.

Theorem 2 (Claim; Lemma 4.2 of the full note). *Under (H1)–(H4) there is $\theta \in [0, \arcsin \eta]$ such that (F, R) is dominated by*

$$\left(\cos \theta \prod_i \rho_i, \quad \theta + \sum_i \chi_i \right).$$

Writing a label as $z = \rho e^{i\chi}$, this reads $z_{\text{out}} = \cos \theta e^{i\theta} \prod_i z_i$. If $\prod_i \rho_i > 0$ one may take $\sin \theta = \|Q\mu(R_1, \dots, R_m)\| / \prod_i \rho_i$.

The analytic core is the following inequality. The question we would like a specialist to answer is whether it holds for arbitrary finite arity, and whether §4 correctly derives Theorem 2 from it.

Theorem 3 (Tensor angle inequality; Theorem 3.1). *Let $u_i, v_i \in H_i$ be unit vectors, $\psi_i = \angle(u_i, v_i)$ and $S = \sum_{i=1}^m \psi_i$. For real a, b with $ab \leq 0$,*

$$\left\| a \bigotimes_i u_i + b \bigotimes_i v_i \right\|_{\pi} \leq |a e^{i \min(S, \pi)} + b|,$$

where $\|\cdot\|_{\pi}$ is the projective tensor norm. For $S \leq \pi$ this is an equality.

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2. TWO ELEMENTARY LEMMAS

Lemma 4 (Realization; Lemma 2.2). *If $\text{Gram}(F, R) \preceq K(\rho, \psi)$, then for a 2-dimensional space X with unit vectors $\hat{F}, \hat{R}$ at angle ψ there is a linear map $B : X \rightarrow H$ with $\|B\| \leq 1$, $B\hat{F} = F$ and $B(\rho\hat{R}) = R$. In particular $\|F\| \leq 1$ and $\|R\| \leq \rho$.*

Proof. The Löwner inequality says $\|\alpha F + \beta R\| \leq \|\alpha\hat{F} + \beta\rho\hat{R}\|$ for all real α, β . So B is well defined and contractive on the span. Extend it by 0 on the orthogonal complement; this also covers $\rho = 0$ and $\psi \in \{0, \pi\}$. $\square$

Lemma 5 (Planar domination; Lemma 2.3). *Let $S \in [0, \pi]$, and suppose $\|af + bg\| \leq |ae^{iS} + b|$ for all real a, b with $ab \leq 0$. Then $\text{Gram}(f, g) \preceq K(1, \psi)$ for some $\psi \in [0, S]$.*

Proof. Let $G = \text{Gram}(f, g)$. The cases $b = 0$ and $a = 0$ give $G_{11} \leq 1$ and $G_{22} \leq 1$. Put $\mu = \sqrt{(1 - G_{11})(1 - G_{22})}$. The hypothesis reads

$$a^2(1 - G_{11}) + b^2(1 - G_{22}) - 2|ab|(\cos S - G_{12}) \geq 0,$$

which holds for all a, b if and only if $G_{12} + \mu \geq \cos S$. Since $K(1, \psi) - G$ has nonnegative diagonal, it is $\succeq 0$ if and only if $\cos \psi \in [G_{12} - \mu, G_{12} + \mu]$. This interval meets $[\cos S, 1]$, because $G_{12} - \mu \leq 1$ and $G_{12} + \mu \geq \cos S$. $\square$

Remark. Only a *lower* bound on the alignment G_{12} is available, and none is needed. Excess alignment is absorbed by choosing $\psi < S$. This is where the argument departs from a naive one that fixes the angle to equal S .

3. PROOF OF THE TENSOR ANGLE INEQUALITY

Reductions. Take $a = 1$ and $b = -t$ with $t \geq 0$. If $S \geq \pi$ the right side is $1 + t$, which is the triangle inequality, so assume $S < \pi$. One-dimensional factors embed isometrically into $\mathbb{R}^2$ without changing projective norms.

Induction on m . The case $m = 1$ is $\|u - tv\|^2 = 1 + t^2 - 2t \cos \psi_1$. Let $m \geq 2$, $X = \bigotimes_{i \geq 2} u_i$, $Y = \bigotimes_{i \geq 2} v_i$ and $S' = S - \psi_1$.

- (1) By associativity of $\otimes_\pi$ and duality, the norm equals $\sup_\beta [\beta(u_1, X) - t\beta(v_1, Y)]$ over bilinear forms β with $|\beta(z, W)| \leq \|z\| \|W\|_\pi$.
- (2) Fix a 2-plane $\Pi \ni u_1, v_1$. Then $\beta(z, W) = \langle z, \Lambda W \rangle$ for $z \in \Pi$, with Λ linear and $\|\Lambda W\| \leq \|W\|_\pi$.
- (3) Put $p = \Lambda X$ and $q = \Lambda Y$. By induction, $\|\alpha p + \beta' q\| \leq |\alpha e^{iS'} + \beta'|$ for $\alpha, \beta' \leq 0$. Lemmas 5 and 4 then give unit $\hat{p}, \hat{q}$ at angle $\tilde{\psi} \leq S'$ and a contraction C with $C\hat{p} = p$ and $C\hat{q} = q$.
- (4) With Rot the rotation of Π taking v_1 to u_1 , $\beta(u_1, X) - t\beta(v_1, Y) = \langle u_1, C\hat{p} - t \text{Rot } C\hat{q} \rangle \leq \|C\hat{p} - t \text{Rot } C\hat{q}\|$.
- (5) Let $V = (C, (I - C^*C)^{1/2})$, an isometry into $\Pi \oplus D$, and $\tilde{R} = \text{Rot} \oplus I$. For unit y we have $\langle y, \tilde{R}y \rangle \geq \cos \psi_1$, hence $\angle(y, \tilde{R}y) \leq \psi_1$. The Π -component of $V\hat{p} - t\tilde{R}V\hat{q}$ is $C\hat{p} - t \text{Rot } C\hat{q}$. The vectors $V\hat{p}$ and $t\tilde{R}V\hat{q}$ have norms 1 and t , and the angle between them is at most $\tilde{\psi} + \psi_1 \leq S < \pi$.

Hence the norm is at most $|1 - te^{iS}|$.

Equality for $S \leq \pi$. Identify each $\Pi_i \cong \mathbb{C}$ with $v_i \mapsto 1$ and $u_i \mapsto e^{i\psi_i}$, and let π_i be the orthogonal projection onto Π_i . For $|\lambda| = 1$, the form $W(x) = \text{Re}(\lambda \prod \pi_i x_i)$ has norm at most 1 and gives $\text{Re}(\lambda(e^{iS} - t))$. Choose λ . $\square$

4. PROOF OF THE CLAIM

Step 0: substitute the children. By Lemma 4, choose unit $\hat{F}_i, \hat{R}_i$ at angles $\psi_i \leq \min(\chi_i, \pi)$ and contractions B_i with $B_i \hat{F}_i = F_i$ and $B_i(\rho_i \hat{R}_i) = R_i$. Put $\mu'(x) := \mu(B_1 x_1, \dots, B_m x_m)$, so $\|\mu'\| \leq 1$, and let $\rho := \prod_i \rho_i$. Then $F = \mu'(\hat{F})$ and $\mu(R_1, \dots) = \rho \mu'(\hat{R})$.

Case $\rho = 0$. Some $R_i = 0$, so $R = 0$ and $\|F\| \leq 1$. Hence $\text{Gram}(F, 0) \preceq K(0, 0)$; take $\theta = 0$.

Case $\rho > 0$.

- (i) *Closure normalization.* By (H4) and $\|R_i\| \leq \rho_i$, $\|Q\mu'(\hat{R})\| = \|Q\mu(R_1, \dots)\|/\rho \leq \eta$.
- (ii) *Raw output satisfies (N-).* Put $f = \mu'(\hat{F})$, $g = \mu'(\hat{R})$, $S = \sum \psi_i$ and $S' = \min(S, \pi)$. For $ab \leq 0$,

$$\|af + bg\| = \sup_{\|\omega\| \leq 1} \langle \omega, \mu'(a \otimes \hat{F} + b \otimes \hat{R}) \rangle \leq \|a \otimes \hat{F} + b \otimes \hat{R}\|_\pi \leq |ae^{iS'} + b|$$

by Theorem 3.

- (iii) *Domination of the raw output.* Lemmas 5 and 4 give unit $\hat{F}', \hat{E}'$ at angle $\tilde{\psi} \leq S'$ and a contraction C with $C\hat{F}' = f$ and $C\hat{E}' = g$.
- (iv) *Dilation.* Let $V = (C, (I - C^*C)^{1/2})$ and $P' = P \oplus I$, and set $\tilde{F} = V\hat{F}'$, $\tilde{R} = \rho P'V\hat{E}'$. The H -components of $\tilde{F}$ and $\tilde{R}$ are F and R , hence $\text{Gram}(F, R) \preceq \text{Gram}(\tilde{F}, \tilde{R})$.
- (v) *Norms and angle.* $\|\tilde{F}\| = 1$. Since $\|(I - P')V\hat{E}'\| = \|Qg\| \leq \eta$, put $\theta := \arcsin\|Qg\|$. Then $\|\tilde{R}\| = \rho \cos \theta$. If $\tilde{R} \neq 0$,

$$\angle(\tilde{F}, \tilde{R}) \leq \angle(V\hat{F}', V\hat{E}') + \angle(V\hat{E}', P'V\hat{E}') = \tilde{\psi} + \theta \leq \sum \chi_i + \theta.$$

Therefore $\text{Gram}(\tilde{F}, \tilde{R}) = K(\rho \cos \theta, \angle(\tilde{F}, \tilde{R}))$. If $\tilde{R} = 0$, the pair is dominated by $K(0, 0)$. $\square$

5. WHY THIS MATTERS AND WHAT HAS BEEN CHECKED

Applied at every non-root node of a projected multilinear tree, Theorem 2 makes the labels $z = \rho e^{ix}$ compose by complex multiplication. As a result, the sharp projected-error constant of such a tree depends only on its number of internal nodes; this is Theorem R of the full note. The same construction in $\mathbb{R}^2 \cong \mathbb{C}$ attains the bound. If Theorem 2 fails, Theorem R falls back to the proved cases: chains of every length, all trees with at most four internal nodes, and bilinear roots fed by two chains.

Numerical controls, which are not part of the proof:

- The claim, at arities 1–3: 344 000 random cases plus adversarial optimization, and 14 000 degenerate cases. No counterexample was found; the largest violation at the proof angle is $6.7 \cdot 10^{-16}$.
- Theorem 3, by cutting-plane LP for $m = 2, 3, 4$: the upper estimate is within $1.6 \cdot 10^{-7}$ of the bound.

Questions for the reviewer. (1) Is Theorem 3 correct for every finite m ? Is it known?

- (2) Does §4 correctly derive Theorem 2? In particular, are Lemma 5 and the dilation step applied under their exact hypotheses?
- (3) Is anything used beyond (H1)–(H4)?